



Høyskolen
Kristiania



Skills

[Take skill quiz](#)

Security

 Endorsed by Frode Hommedal and 2 others who are highly skilled at this Endorsed by 2 colleagues at Equinor 25 endorsements

Antivirus

 Endorsed by Carl Bretteville and 2 others who are highly skilled at this Endorsed by 2 colleagues at Capgemini 19 endorsements

Computer Security

Trond Selland has given an endorsement for this skill

 28 endorsements[Show all 4 skills →](#)

Experience

**Equinor**
Full-time · 2 yrs
Oslo, Norway

- **Advisor Information Security**
Apr 2022 - Present · 4 mos
Responsible for follow up, challenge and advice on information security related topics in the business areas Renewables (REN).
- **Information Security Analyst**
Aug 2020 - Jul 2022 · 2 yrs

**Managing Consultant, Cyber Security**
Capgemini
Feb 2019 - Aug 2020 · 1 yr 7 mos
Oslo Area, Norway**Security Analyst**
Statoil
Nov 2011 - Feb 2019 · 7 yrs 4 mos
Stavanger Area, Norway

Henrik Ramberg
Information Security Advisor at Equinor
Oslo, Oslo, Norway · [Contact info](#)

 Equinor
 University of Sunderland

332 connections

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Education

**University of Sunderland**
Master, IT; Software Engineering
2001 - 2002**Gjøvik University College (HiG)**
Bachelor, Datateknikk
1998 - 2001

Course agenda

- Session 1 : Computers and digital technology
- Session 2: Network and data communication
- Session 3: Global communication – internet
- Session 4: Practical examples & demos
- Home work: 2x labs from TK 1100

Session 1: Computers and digital technology

Session 1: Computers and digital technology

What is a computer?

- General use
 - The computer is used in many different areas
- Electronic
 - The computer needs electricity to function.
- Digital
 - The computer is based on binary logic circuits (everything is **represented** as **numbers**)
- Human-layer
 - The computer is designed by humans to produce results that humans are interested in.

Computer applications



This is a
computer



.. also a
computer



Leonardo
(supercomputer)



Apple smart
watch



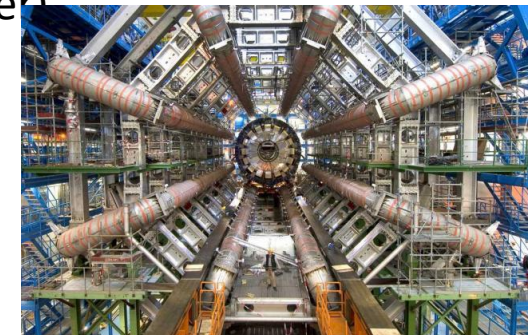
Smart
card



Coffee
machine
with
wifi...



Fireplace with wifi and
Alexia...



Particle
accelerator

Digitalisation

- In a computer **everything is represented as numbers**
 - Numbers, letters, pictures, sound, film, instructions...

The image is of an audio file, it is just a long number that is interpreted in relation to a pre-defined format...

Documents and Settings\blistog\Mine dokumenter\Min musikk\NiTimen.mp3															
	0001	0203	0405	0607	0809	0A0B	0C0D	0E0F	0123456789ABCDEF						
0x1920	10B4	473A	115A	FD5D	D2C8	CC5B	839C	A879	. 'G: .Zý] ÒÈÌ[fœ`y						
0x1930	116D	4442	0C2F	267A	95EF	0FEB	0400	3B61	.mDB./ &z•i.ë..;a						
0x1940	5CE5	0E14	68AC	9A96	D963	CA0B	273D	F35D	\â..h¬š-ÙcÊ.'=ó]						
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0x1960	BA21	2383	744B	7CBC	D4D0	E204	2DC7	CF2E	°!#ftK ¼ÔBâ.-çİ.						
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0x19E0	6D87	3EC4	A12D	C93A	AB68	A891	95B9	FFFE	m‡>Ä¡-É:«h``·¹yb						
0x19F0	DEAF	BF66	A9C1	BD18	7C31	17D1	021B	406B	E`¿f@Á½. 1.Ñ..@k						
0x1A00	FFDF	A32B	FFDA	2018	7840	A868	5425	1F0D	ýßž+ýÚ .x0`hT%..						
0x1A10	9923	4A64	8DC9	E046	358B	3378	3ADD	0C5A	™#Jd□ÉàF5<3x:Ý.Z						
0x1A20	1061	5DC4	3E69	11A8	E810	ACF3	4D52	D25C	.a]Ä>i..`è.-óMRÒ\						
0x1A30	DE2D	00A1	064C	6456	F8BE	7329	9219	8E34	E-.¡.LdVø¾s)' .Ž4						

Number systems - decimal numbers

- Decimal numbers are based on the number of fingers (toes)
- Dozens, tens, and other older ways of counting
- Roman numerals
 - MCMLXXIV = 1974 (I=1, V=5, X=10, L=50, C=100, M=1000)
- 0 invented around 600 AD (Indian numerals)
- Introduced to Europe by Leonardo of Pisa (Fibonacci) around 1200

$$\begin{aligned} 1904 &= \mathbf{1} * 1000 + \mathbf{9} * 100 + \mathbf{0} * 10 + \mathbf{4} * 1 \\ &= \mathbf{1} * 10^3 + \mathbf{9} * 10^2 + \mathbf{0} * 10^1 + \mathbf{4} * 10^0 \end{aligned}$$

Number systems - binary numbers

- Based on two states (digital; digit = finger or toe)
 - finger up, finger down
 - off, on (low signal, high signal)
 - 0, 1

$$25_{10} = 11001_2 = 1 * 16 + 1 * 8 + 0 * 4 + 0 * 2 + 1 * 1$$
$$= 1 * 2^4 + 1 * 2^3 + 0 * 2^2 + 0 * 2^1 + 1 * 2^0$$

- Will quickly become a lot of digits
- A binary digit is called a bit. (binary digit)(b)
- A group of 8 binary digits is called a byte (B)

Counting with **binary numbers**

0 01 10 11

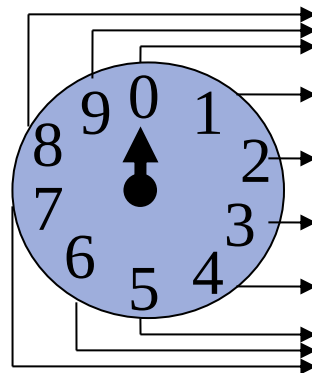
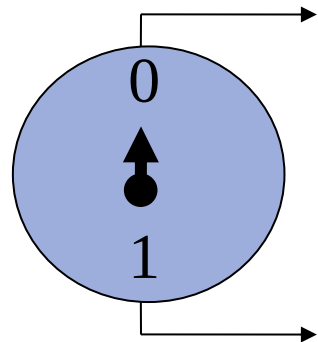
100 101 110 111

1000 1001 1010 1011

1100 1101 1110 1111

Why do computers use binary logic?

- The binary system is simple and reliable
- Simple circuits make cheap circuits
- Greater complexity in basic building blocks increases the likelihood of errors
- It is the (theoretically) most efficient symbol system in existence!



Conversion from decimal to binary

$2^0 = 1$	$851 =$	$512 + 339 =$	$2^9 + 339$
$2^1 = 2$	$339 =$	$256 + 83 =$	$2^8 + 83$
$2^2 = 4$	$83 =$	$64 + 19 =$	$2^6 + 19$
$2^3 = 8$	$19 =$	$16 + 3 =$	$2^4 + 3$
$2^4 = 16$	$3 =$	$2 + 1 =$	$2^1 + 1$
$2^5 = 32$	$1 =$		2^0

$2^7 = 128$
 $2^8 = 256$

$851 = 2^9 + 2^8 + 2^6 + 2^4 + 2^1 + 2^0$

$2^9 = 512$
 $2^{10} = 1024$

$851 = 1 \cdot 2^9 + 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$

$851_{10} = 0000\ 0011\ 0101\ 0011_2$

Binary to decimal conversion

$$\begin{aligned} 2^0 &= 1 & 1101010011 &= 1*2^9 + 1*2^8 + 0*2^7 + 1*2^6 + 0*2^5 + 1*2^4 + 0*2^3 + 0*2^2 + 1*2^1 + 1*2^0 \\ 2^1 &= 2 & & \\ 2^2 &= 4 & & \\ 2^3 &= 8 & & \\ 2^4 &= 16 & & \\ 2^5 &= 32 & & \\ 2^6 &= 64 & & \\ 2^7 &= 128 & & \\ 2^8 &= 256 & & \\ 2^9 &= 512 & & \\ 2^{10} &= 1024 & & \end{aligned}$$

512 256 128 64 32 16 8 4 2 1

1101010011

512 256 64 16 2 1

NB! Leading zeros are missing in 16 bit precision!

0000 0011 0101 0011

Hexadecimal numbers

- Using **16** as base
 - Then have to invent 6 "new" digits
 - $A_{16}=10, B_{16}=11, C_{16}=12, D_{16}=13, E_{16}=14, F_{16}=15$
 - **0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F**
 - $11_{16} = 1*16 + 1*1 = 17_{10}$
- $$0A4C_{16} = 0*16^3 + 10*16^2 + 4*16^1 + 12*16^0$$
- $$0*4096 + 10*256 + 4*16 + 12*1 = 2636_{10}$$
- Translates from binary to hex by grouping 4 and 4 binary numbers together (“nibbles”)
 - **1010 0011 1001 1111**₂ = 0x **A 3 9 F**₁₆
 - $1010_2 = A_{16}, 0011_2 = 3_{16}, 1001_2 = 9_{16}, 1111_2 = F_{16}$

ASCII Table (7 bit)

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

- "A" = 0x41 -> 0100 0001
"a" = 0x61 -> 01 **1** 0 0001
 - A bit of difference between uppercase and lowercase letters
- The numbers **are encoded** with: 0x30-0x39
 - The binary number 0000 0000 is in ASCII: 0011 0000
- Space is 0x20
- Line feed is CarriageReturn LineFeed (0x0D 0x0A)

There are 10 types of people in the world,
those who can understand binary and those who
can't.

Boolean algebra

	y	
$\wedge$	0	1
x	0	0
	1	1

	y	
$\vee$	0	1
x	0	1
	1	1

	y	
$\rightarrow$	0	1
x	0	1
	1	1

	y	
$\oplus$	0	1
x	0	1
	1	0

Figure 1. Truth tables



Figure 2. Logic gates



Figure 3. De Morgan equivalents



Figure 4. Venn diagrams

Boolean algebra

	y	
$\wedge$	0	1
x	0	0
	1	1

	y	
$\vee$	0	1
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	1	1

	y	
$\rightarrow$	0	1
x	0	1
	1	1

	y	
$\oplus$	0	1
x	0	1
	1	0

Figure 1. Truth tables



Figure 2. Logic gates



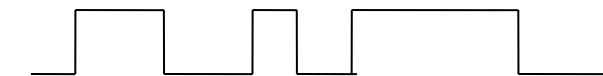
Figure 3. De Morgan equivalents



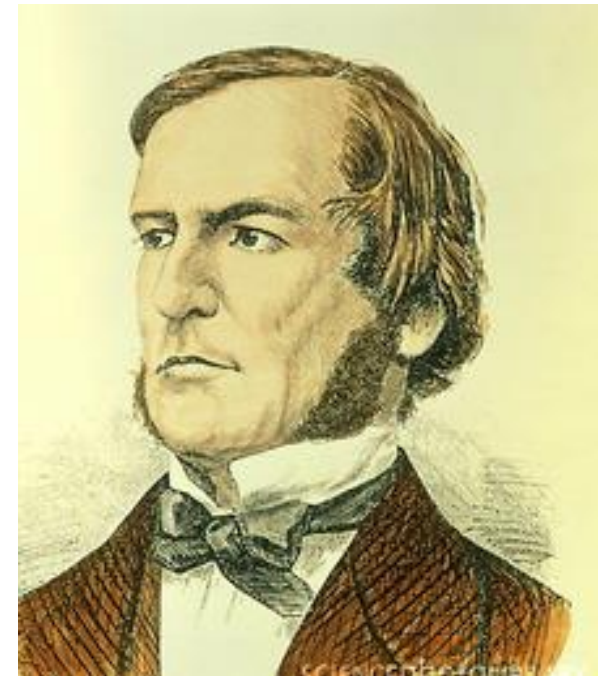
Figure 4. Venn diagrams

Bit and George Boole

- All logic in **digital** electronics is based on **bits**.
- Using transistors as building blocks
- Can be 0 or 1; high or low voltage
- The logic used is called **Boolean Algebra**.
 - **George Boole** (1815-1864)

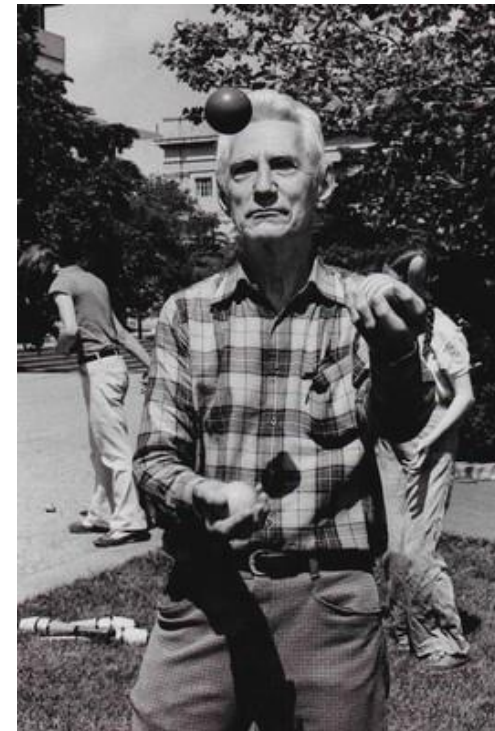


0 1 1 0 0 1 0 1 1 1 1 0



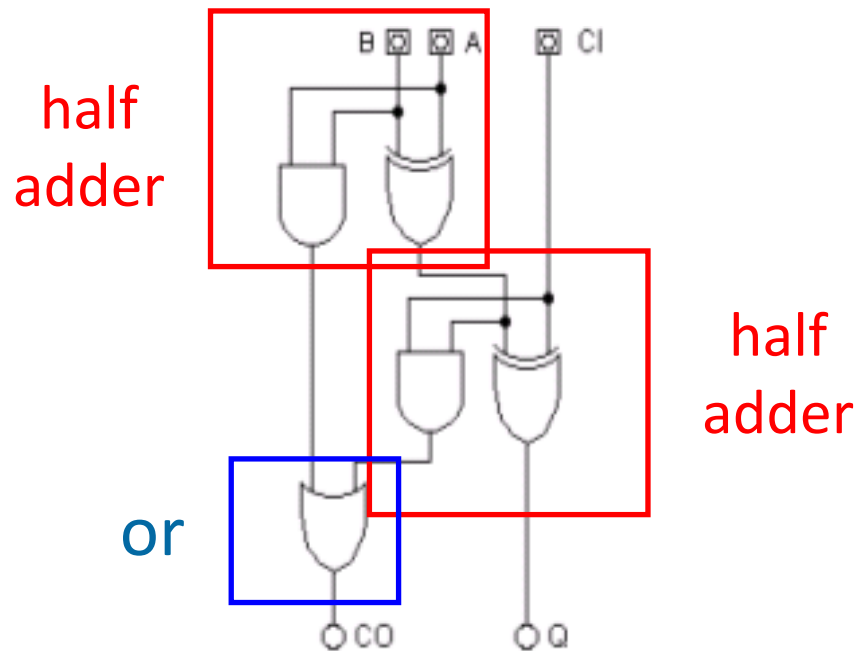
Boolean algebra

- Consists of 3 (4) basic operations (gates)
 - NOT (NOT)
 - AND (AND)
 - OR
 - (EXCLUSIVE OR (XOR))
- "Rediscovered" in 1939 by Claude Elwood Shannon
- Computer electronics most often prefers "negative" logic
 - NAND, NOR, XNOR



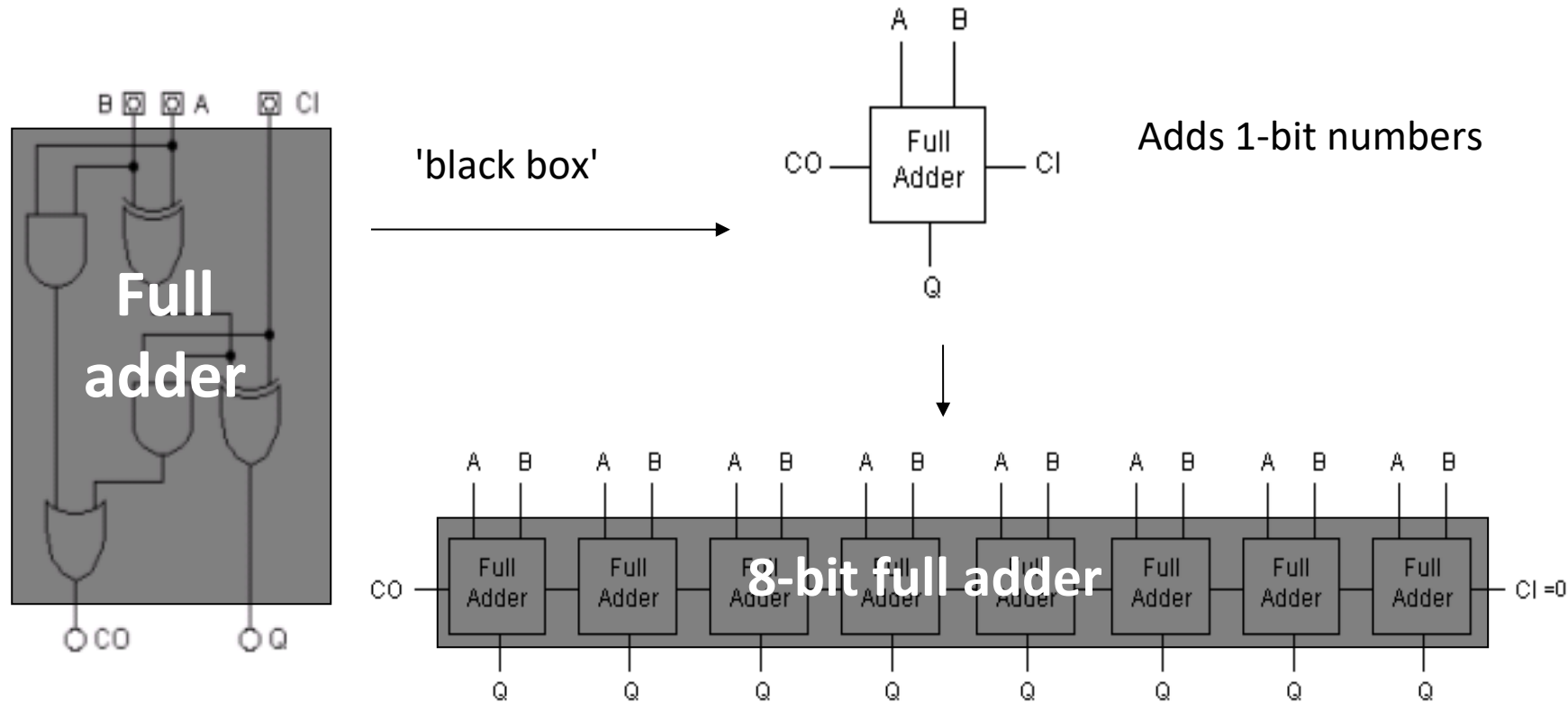
Practical example: Full adder

- Adds two bits, **A** and **B**, and any meaning on **CI**
- Get the answer in **Q** with the meaning out in **CO**



A	B	CI	Q	CO
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Practical example: 8 bit full adder

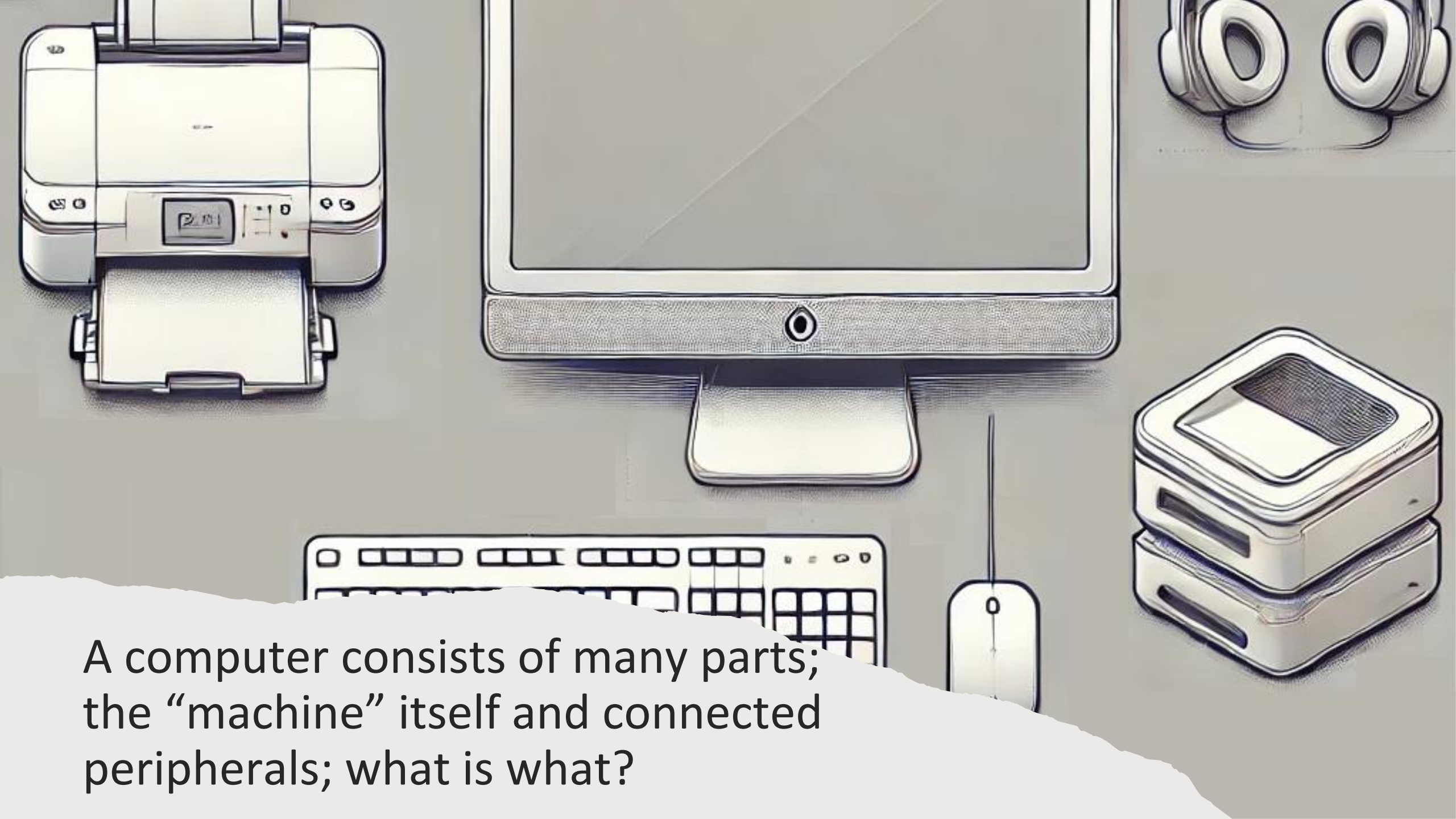


8 full adders connected together: Adds 8 bit numbers together (called a bar adder, and is a bit "slow"...)

Units and sizes

- **Bit** (b) - 0 or 1
- **Byte** (B) = 8 bits
- Kilo = $10^3 = 1000 \approx 1024 = 2^{10}$
 - 1 km = 1000 m, 1 mm = 1/1000 m
- For the sake of simplicity, we'll "cheat" a little!
 - k = 1000, Ki = 1024, (assume K = 1024 as well)
- Kilobyte/ **KibiByte** (KiB) = 2^{10} bytes = 1024 bytes
- Megabyte/ **MibiByte** (MiB) = 2^{20} bytes = 1024 KB = 1048576 bytes
- Gigabyte/ **GibiByte** (GiB) = 2^{30} bytes = 1024 MB = 1073741824 bytes
- **Herz** (Hz) = events per second
- **MIPS** = Mega instructions per second.
- **FLOPS** = Floating point operations per second
- **kbps** = 1000 bit/second (bit rate, "bandwidth")

Multiples of bytes v · d · e				
SI decimal prefixes		Binary usage	IEC binary prefixes	
Name (Symbol)	Value		Name (Symbol)	Value
kilobyte (kB)	10^3	2^{10}	kibibyte (KiB)	2^{10}
megabyte (MB)	10^6	2^{20}	mebibyte (MiB)	2^{20}
gigabyte (GB)	10^9	2^{30}	gibibyte (GiB)	2^{30}
terabyte (TB)	10^{12}	2^{40}	tebibyte (TiB)	2^{40}
petabyte (PB)	10^{15}	2^{50}	pebibyte (PiB)	2^{50}
exabyte (EB)	10^{18}	2^{60}	exbibyte (EiB)	2^{60}
zettabyte (ZB)	10^{21}	2^{70}	zebibyte (ZiB)	2^{70}
yottabyte (YB)	10^{24}	2^{80}	yobibyte (YiB)	2^{80}



A computer consists of many parts;
the “machine” itself and connected
peripherals; what is what?

Computer - main parts

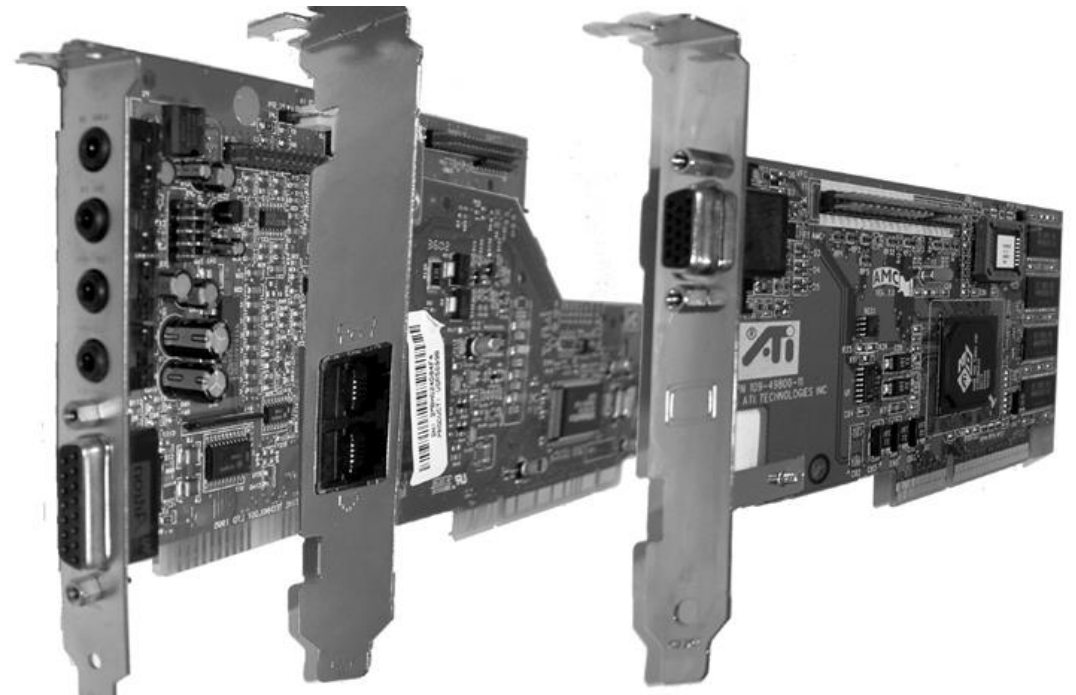
- The computer system can be divided into 2 main parts
- Hardware
 - *Electronic* and *electro-mechanical* components that make the computing environment work
 - Input/Output, Storage (HD, RAM), Computing (CPU, GPU)
- Software
 - **Applications** that exploit the capabilities provided by the hardware to perform special tasks
 - Applications are **programs** that are written in a language that the computer "understands"

I/O Hardware

- Almost endless choice of I/O devices
- Common concepts
 - Port
 - Bus (*daisy chain or shared direct access*)
- Controller (*controller or host adapter*)
- I/O instructions control devices
- Devices have addresses, which are used by
 - Direct I/O instructions
 - Memory-mapped I/ O
- The interaction with the OS is most often done through a **driver** that translates OS instructions into Controller instructions.

Expansion Card - PCI

- Cards with their own circuits can be connected to the motherboard
- There are many standards,
 - PCI(e) and AGP are the most common
- Graphics card
- Video editing card



Input data units

- Standard keyboard is 104-key IBM PC standard (qwerty)
- Mice today are optical
 - LED or LED laser emits a beam of light
 - Low-resolution "camera" (20x20 px ->) takes "pictures" approx.
every 1/10 second (often more often)
 - Chip measures/calculates difference between images and calculates position change



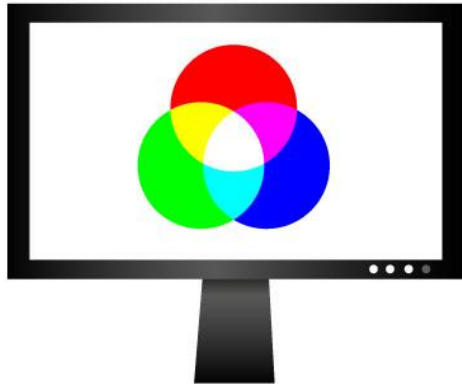
Out data units

- Screen, showing pixels on a glass surface
- Printer, draws pixels on a sheet of paper

RGB

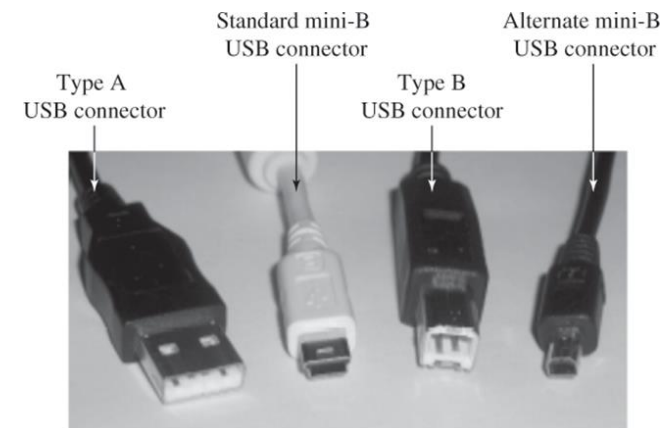
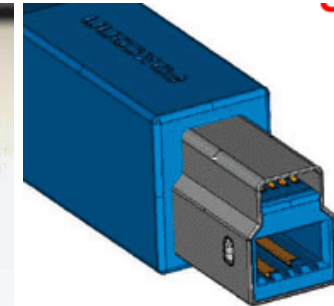
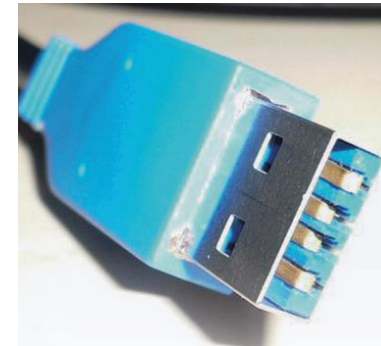
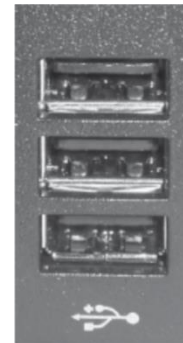


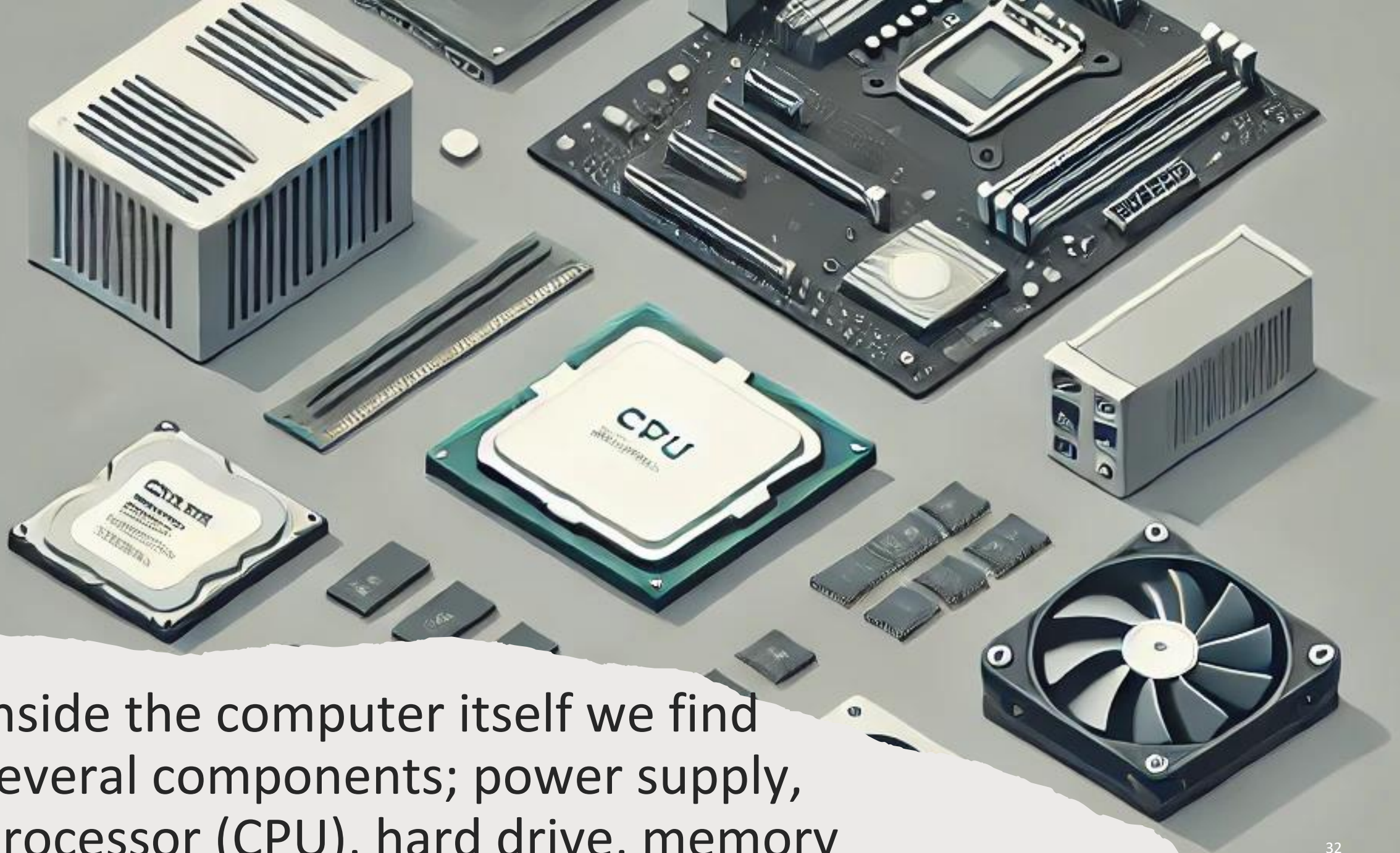
CMYK



USB – Universal Serial B us

- Supports up to 127 different connected devices
 - Expanded with **hub**
- High-speed data transfer
 - USB 1.1: 12Mbps (max)
 - USB 2.0: 480Mbps (max)
 - USB 3.0: 5Gbps (max)
- Variable bitrate
 - Low-speed channels for slow equipment
- Four main types of data transfer
 - Bulk, interrupted, isochronic, control
- PnP and hot-pluggable
- Can be used for charging

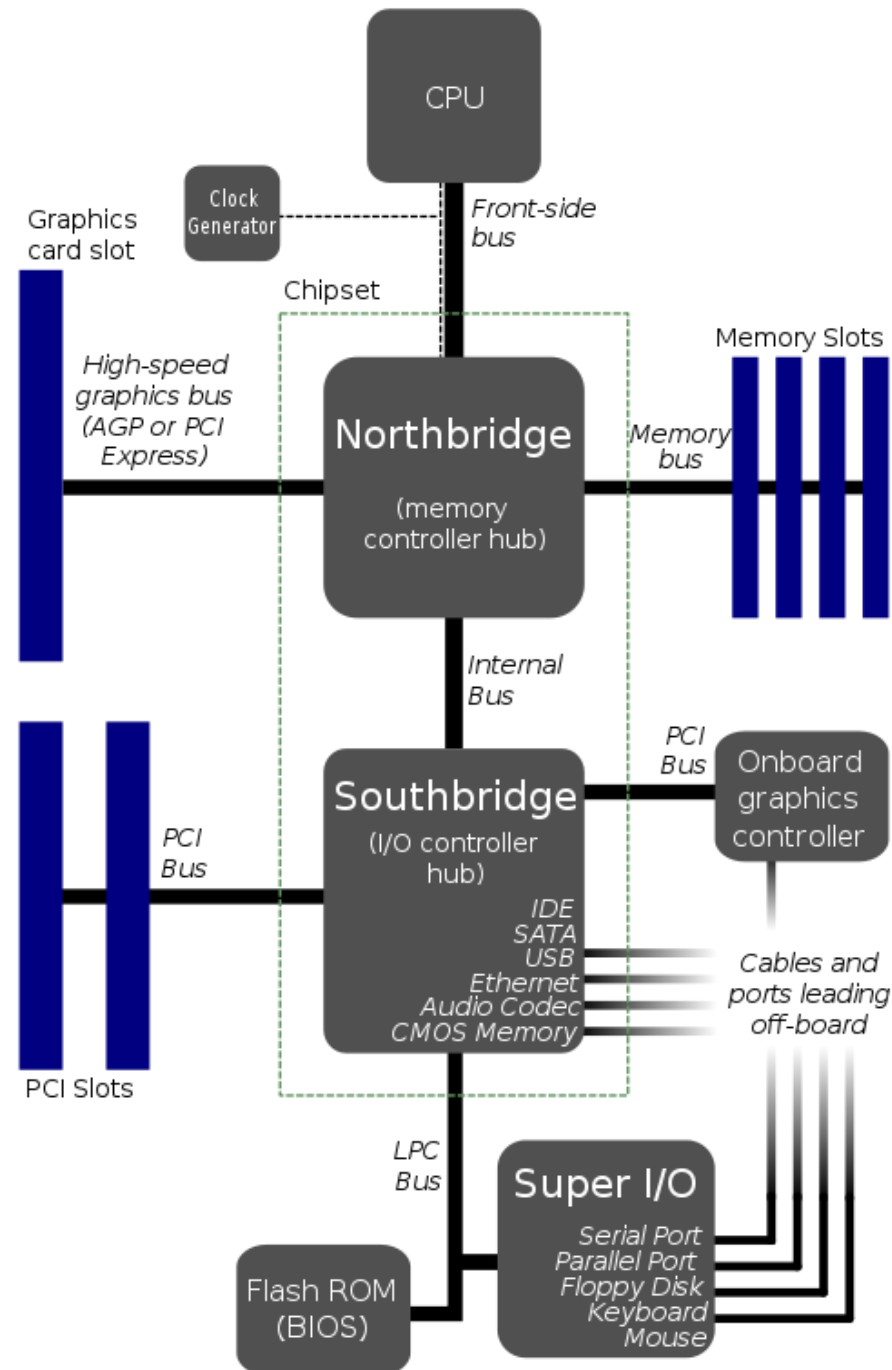




Inside the computer itself we find several components; power supply, processor (CPU), hard drive, memory

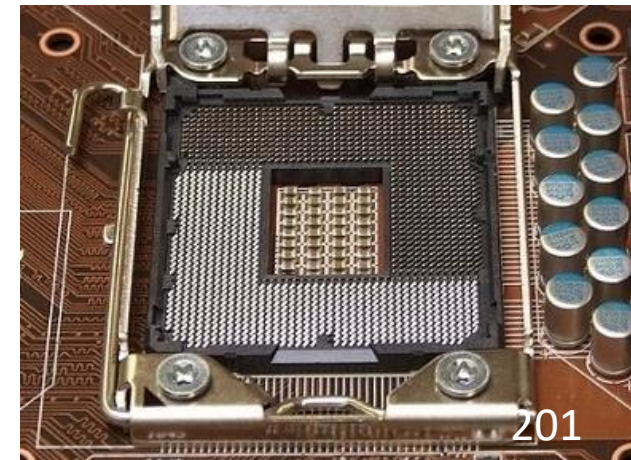
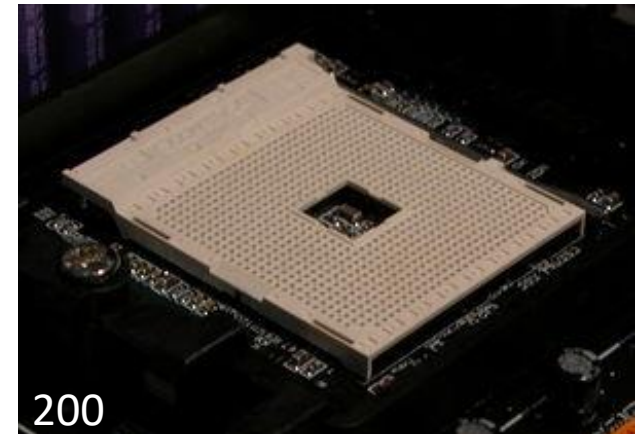
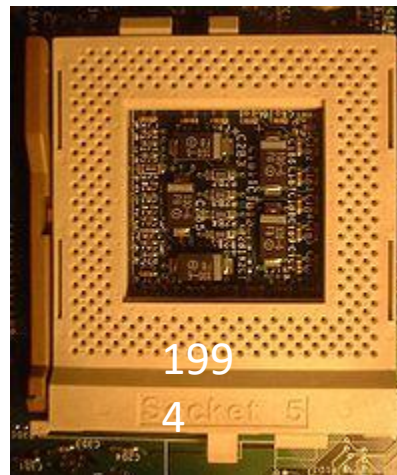
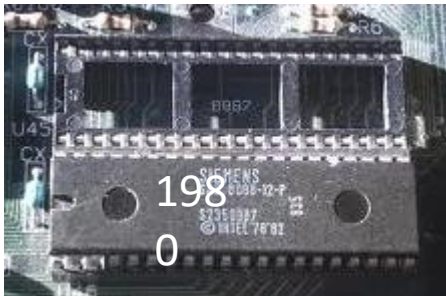
Motherboard

- Connects:
 - CPU
 - Chipset
 - Buses
 - Memory slots and RAM
 - Expansion slots and add-in cards
 - Ports and "plugs"



Processor (CPU)

- CPU - Central Processing Unit - is the "brain" itself
- The CPU executes all instructions in the software
- The CPU is connected to the motherboard through a **socket**.
- Different types with different:
 - number of CPU pins
 - FSB (Front Side Bus) Speed
- Depends on motherboard



Memory (RAM)

- RAM – Random Access Memory – is the memory, or storage, of a computer
 - BIT
 - **Binary** Dig **IT**
 - On/off circuit
 - 1 or 0
 - BYTE
 - 8 BIT
 - Can store, for example, an alphanumeric character
 - Smallest addressable unit in RAM

Mass storage devices

- Used for permanent storage of large amounts of data
 - Punch card
 - Punch tape
 - Floppy disk, Flash memory
 - Zip, Jazz
 - Disk
 - CDs, DVDs
 - Magnetic tape (reel tape, cassette)
- Transmission to the processor via a controller
 - IDE/EIDE, SATA, SCSI, ...



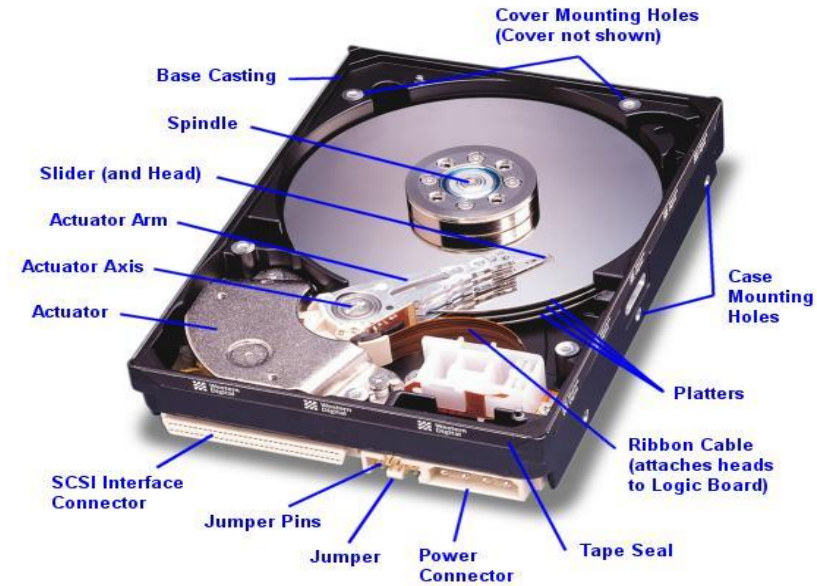
Development, increased storage space



Punched cards and paper strips (1804 – 1980)



"Floppy" disk (1970 – 2000)



Modern hard drive (1960–present)



DVD – flash drive – SSD M.2 disk (today)

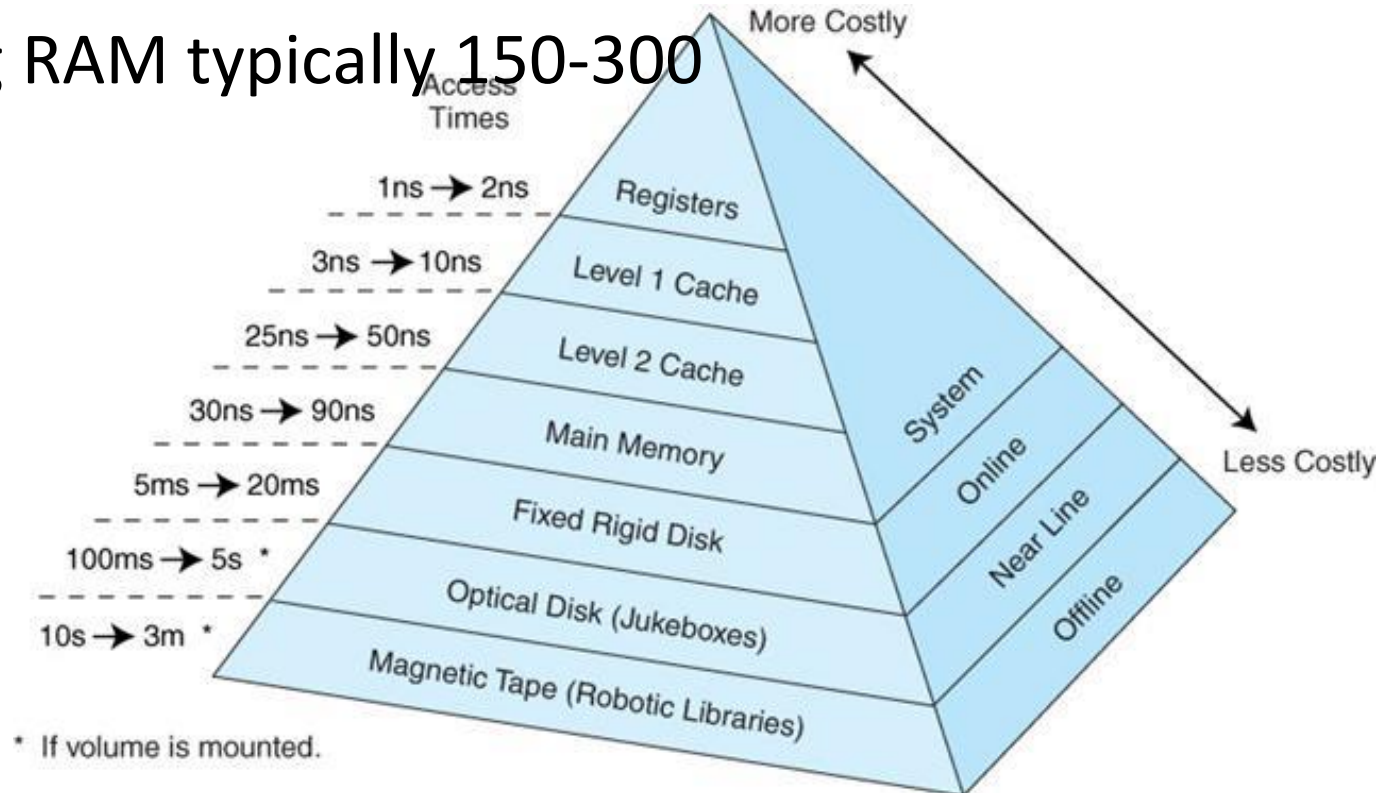


From CPU to computer programs

- **Instructions to the processor = programming**
- Retrieved into the processor from memory (RAM)
- Machine language (IA32, IA64, etc.)
- Assembly language (each command corresponds to a machine instruction:
`mov eax, [00AF3B13] )`
- Low-level language
 - C, C++ (and more)
 - Compiled into machine instructions
- High-level language
 - C#, JAVA (and many others)
 - Interpreted in an environment (e.g. a virtual machine)
- Other “languages”
 - Excel, SQL, scripting languages

The memory hierarchy

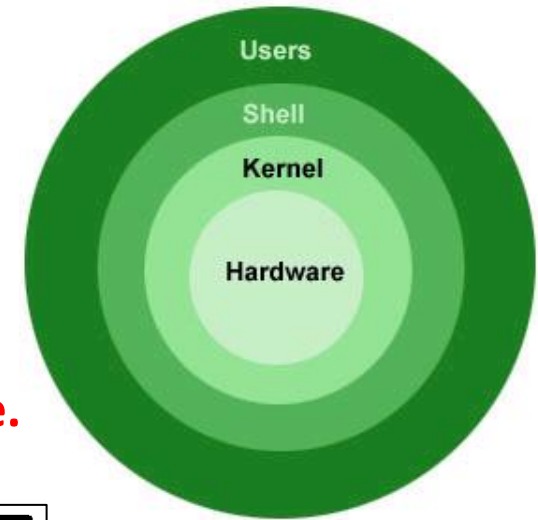
- Doing something in a register typically takes one clock cycle
- Reading/writing RAM typically 150-300



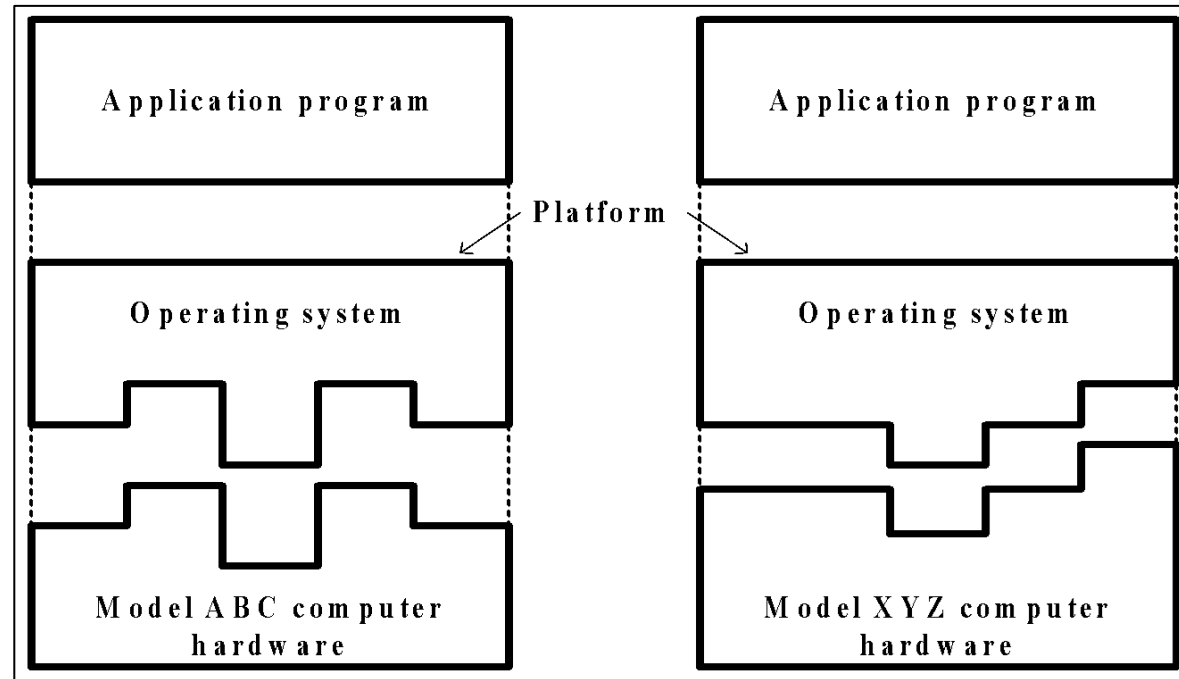
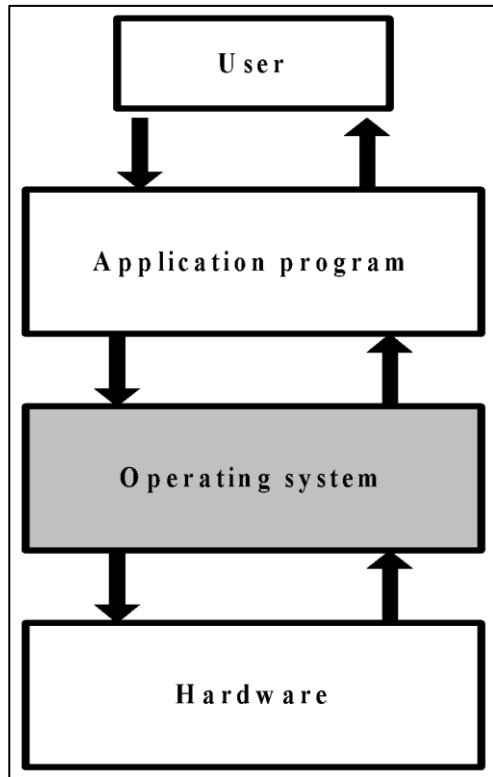
Cache

- The locality principle
 - About 5% of the code is executed about 95% of the time
 - Instructions to be executed are stored in the order in which they are to be executed
- Caching takes advantage of this by temporarily storing the most recently read instructions in cache memory.
 - Lots of loops mean easy accessibility...
- + Paging
 - We typically do not read individual instructions into Cache, but memory blocks of 4 KiB.

What is an operating system?



The operating system is software that sits between the user and the hardware.



Same application on different types
hardware (OS = “platform”)

What is an operating system?

- It is an **extended/abstract machine** (top-down view)
 - Hides the “gritty” details in HW
 - Offers the user and programmer a **virtual machine** that is easier to program/use
- It is a **resource manager** (bottom-up view)
 - Each program gets **time** on the resource
 - Each program gets **space** on the resources (CPU, Memory,...)
- **Network** management , **file** organization , **security**

Abstraction

- To abstract means to remove (unimportant) details.
- For example: A **file** will (most often) consist of metadata (data about data)
 - A name
 - when created, last modified, etc.
 - how big (KiB)
 - which sectors on the disk the data is physically located on
 - ... and the content itself (data/instructions)
- The files are organized in **directories** / ~~folders~~.
 - another abstraction, which is also **a type of file** that contains metadata and addresses to other files
- **Physically** /concretely, it is just magnetized areas in addressed sectors on a disk/CD/memory stick.
- To the user, this abstraction appears as symbols he/she can click on.
- For the programmer, the OS offers **system calls** that allow him/her to create, open, close, position himself/herself within, read from, and write to – these files.

Data program

- All programs are (today) made by humans
 - Desktop software; Word, Excel
 - Web/Cloud software; Elements, DFØ
 - Mobile "apps"; CandyCrush
- These run on a CPU, and use the machine's resources such as memory, disk, screen, keyboard, etc.
- And as we will discuss in later lectures, they can contain errors that lead to security vulnerabilities...

PC vs Server vs Virtualization

- A PC – Personal Computer – is what we have come to understand today; a machine with a screen and keyboard.
- A Server can be seen as a more powerful version
 - More and more powerful CPUs
 - More memory
 - Often does not have a monitor and keyboard connection
 - May not have a graphical interface, only text-based
- A "virtual" machine has hardware simulated in software :-)



Seriously – 42 slides?

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Answer to the Ultimate Question of Life, The Universe, and Everything? "Deep Thought" takes 7.5 million years to answer: 42...



Høyskolen
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